

System Architecture and Mission Feasibility of a Multirotor Communications Relay

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Abstract—An airborne relay can restore a radio path across an obstruction, but the aircraft must carry and power the payload long enough to provide useful service. This paper examines which placement and dwell targets warrant further development for a notional multirotor relay. The architecture assigns mission traffic to the payload and aircraft command, electrical power, and mechanical support to the carrier. That allocation defines two screens: a link check on both relay hops and a carrier check that closes the battery, mass, and hover-power loop. At 10 km separation and 120 m relay height, the reference 30-minute case closes at 4.77 kg with a 0.50 m rotor, while the 45-minute case exceeds both size limits. Moving the relay toward the ground endpoint raises the weaker-hop margin by about 1 dB without changing modeled hover demand, which makes it a provisional candidate for site assessment. Published hover data constrain one power coefficient. Fitting it to each test point alone moves the rotor-size boundary from 33 to 49 minutes, and pairing the least favorable point with a 15 percent hover-power allowance places that boundary near 28 minutes. The 30-minute target is therefore a candidate that installed hover-power measurements must confirm. Site-specific link measurements and a payload-fault check address the other risks to that decision. The assessment covers outbound traffic during hover. The contribution is a traceable path, organized by the NASA system design processes, from each failed screen to the design decision and interface test it calls for.

TABLE OF CONTENTS

1. INTRODUCTION.....	1
2. SYSTEM AND ASSESSMENT BOUNDARY	2
3. METHOD.....	4
4. RESULTS.....	9
5. DISCUSSION.....	12
6. CONCLUSIONS.....	13
APPENDICES.....	13
A. COMPLETE CARRIER MASS BALANCE.....	13
B. HOVER CALIBRATION SENSITIVITY	13
DATA AVAILABILITY.....	14
ACKNOWLEDGEMENTS	14
REFERENCES.....	14
BIOGRAPHY	15

1. INTRODUCTION

Consider an operator at a ground control station who is tracking a remote aircraft on a map but has lost line of sight to it because an obstruction sits on the direct path. The operator does not need a stronger radio as much as a better vantage point. A relay aircraft provides one, much as a retransmission team does for a ground unit. Placing a radio above the obstruction can provide two clear paths, one to each endpoint. Research on unmanned aircraft systems (UAS) has examined this opportunity through aerial networking and mobile relaying [1], [2], [3]. The aircraft must sustain the useful geometry while carrying and powering the radio. This makes the communications requirement an aircraft resource requirement as well.

The hard part is keeping the relay there. Suppose the two endpoints are 10 km apart, the remote aircraft flies at 100 m, and an 80 m screen stands between them. A midpoint relay at 120 m clears the screen in the declared geometry. Whether it can remain there for 30, 45, or 60 minutes depends on a different calculation. More battery adds mass, greater mass requires more hover power, and that power requires additional battery. A consistent mass balance can lie within a small-aircraft envelope, lie outside it, or have no finite solution under the assumed scaling relations.

The design question is what to carry forward in the tradespace: a different relay position, a longer dwell, or a different carrier. The architecture assigns mission traffic to the payload and aircraft command, electrical power, and mechanical support to the carrier. These responsibilities establish what each analysis must check. The study first asks whether both radio hops pass, then whether a consistent aircraft can sustain the requested dwell within the size limits. It uses the results to recommend a candidate for further assessment and identify what could overturn that recommendation. Section 2 defines the service and system boundary, following the NASA system design processes shown in Figure 2. Section 3 presents the models and evidence, Section 4 compares the resulting choices, and Sections 5 and 6 identify the next decisions and tests. The appendices retain the derivation and calibration details.

Related Work and Contribution

Communication and aircraft sizing research already supplies the pieces used here. Zeng, Zhang, and Lim optimized relay

transmit power together with relay trajectory [3]. Zeng, Xu, and Zhang combined rotary-wing propulsion energy with communication energy under throughput constraints [4]. Yan et al. treated battery weight as a design variable that trades stored energy against the power needed to lift it [5]. Rodrigues et al. placed relay positioning inside a propulsion-aware energy model [6]. On the aircraft side, Russell, Theodore, and Sekula built conceptual sizing models for small quadcopters from test data [7]. They reported that their sizing did not close for hover missions beyond a certain length. Bershadsky, Haviland, and Johnson matched motors, propellers, and batteries to mission requirements [8]. Table 1 places these alongside the present study.

Table 1. Related work and its relationship to this study

Reference	Emphasis	Relationship
Zeng et al., 2016 [3]	Relay power and trajectory optimization	Basis for relay placement and power
Zeng et al., 2019 [4]	Propulsion and communication energy together	Basis for combined energy reasoning
Yan et al., 2021 [5]	Battery weight and available energy	Basis for the battery mass trade
Rodrigues et al., 2022 [6]	Propulsion-aware relay positioning	Basis for endurance-aware placement
Russell et al., 2018 [7]	Small quadcopter sizing from test data	Basis for sizing feedback and closure limits
Present study	Placement and dwell screening with a hover-power fit	Allocated screens traced to design decisions

This paper makes three primary contributions. First, it states an explicit aircraft-payload allocation of four flows and uses that allocation to decide which analysis or test each flow requires. Second, it pairs a two-hop link screen with a carrier screen that closes battery, mass, and hover power, and it compares placement and dwell choices against both. Third, it retains both failure flags for every case and maps each failed screen to a design decision and to the interface or installed test that could overturn the recommendation. In this paper, system architecture means this bounded allocation and its assessment. The link and sizing relations are standard, so the contribution lies in how the architecture connects them. The four system design processes of the NASA Systems Engineering Handbook [9], shown in Figure 2, organize the trace from service need to criteria, allocation, and candidate design. This trace matters because it shows which evidence would change the decision.

The research question is whether the declared 10 km, 30-minute outbound service target merits further development, and whether placement or carrier changes should come first. The 10 km, 30-minute case serves as the threshold, and the 20 km separation and 45-minute dwell objectives provide more demanding comparison points. The case study is analyst-defined and precedes hardware. Its quantitative scope is one-way radio-path screening during hover.

2. SYSTEM AND ASSESSMENT BOUNDARY

This section carries out the Stakeholder Expectations and Logical Decomposition processes of Figure 2 for the relay. The relay aircraft carries a communications payload and holds an assigned position. In Figure 1(a), traffic travels from ground endpoint G through the payload to remote aircraft U. The payload is treated as a black box in this assessment. Its size, weight, and power (SWaP) allocation is 0.20 kg of mass and 14 W of direct current (DC) power, and the link model gives it a fixed radio frequency (RF) output. The carrier supplies the airframe, propulsion, battery and power distribution, avionics, command radio, and mount. Operators, ground equipment, the remote aircraft, and spectrum authorities are external to the relay system.

Only the time on station is assessed quantitatively. A full mission profile would run from preparation through launch, transit, establishing position, relaying, monitoring, and recovery. This paper calculates the relay interval alone, and only in the direction from G through the relay R to U. The battery reserve holds energy back from that interval.

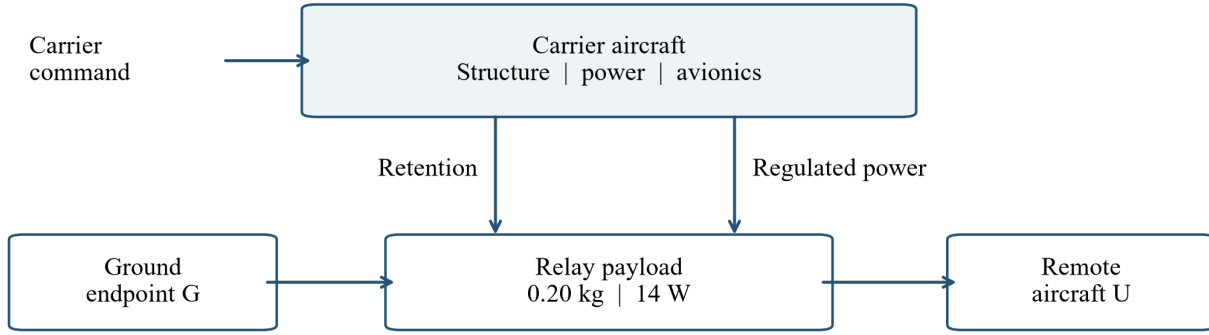
Aircraft-Payload Boundary

Four flows organize the system responsibilities, but only electrical power and mechanical support physically cross the carrier-payload interface. Mission traffic passes through the payload; aircraft command stays on the carrier path. Their failure consequences differ, and so does the evidence each one needs. A blocked mission path loses service and is examined by the link screen. A power shortfall limits dwell and is examined by the carrier screen, which also carries the payload and mount masses. A command-path failure affects control, and an inadequate mount prevents safe support; these two are assessed by interface test rather than calculation. Table 2 links each flow's failure consequence to the evidence needed before the aircraft can be integrated.

Mission traffic. Traffic passes from G through the relay payload to U, as Figure 1(a) shows, so the payload owns this flow. The carrier only has to place the payload at the required position and keep it there. The link check establishes that the declared geometry and RF assumptions give two clear hops with positive margin at some sampled points. However, protocol, waveform, and compatibility with the ground radio and remote aircraft belong to the radio selection that follows. Data rate and a delivery criterion are set with it, and an installed two-hop communication test then verifies the flow.

Aircraft command. The operator commands the carrier and receives its health data over a path separate from the relayed traffic. The intent is to retain aircraft control if the payload fails. Sharing a battery and airframe means that logical separation alone is not fault isolation. A payload overcurrent, for example, could disturb the carrier supply unless the power distribution prevents it. A protected payload branch is therefore a candidate design response. Its assessment would monitor carrier voltage, command continuity, and fault indication during a safely bounded payload-load fault.

(a) Aircraft responsibilities



(b) Worked geometry at 10 km separation

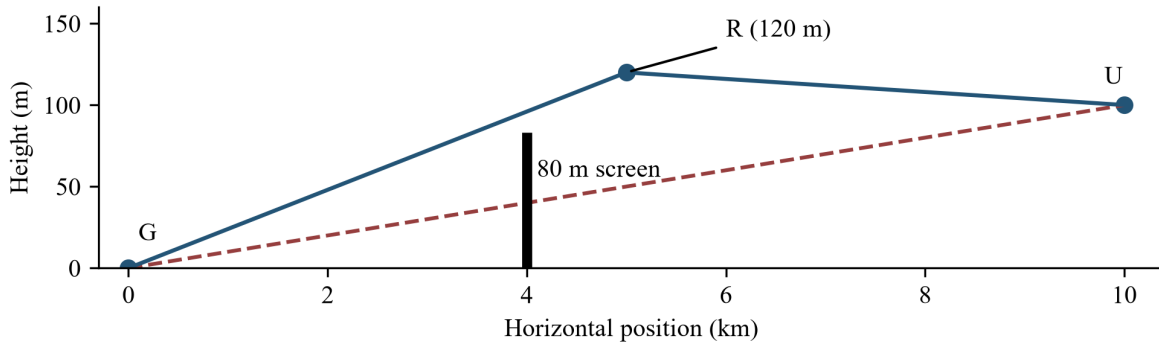


Figure 1. Relay architecture and the worked geometry. (a) The aircraft supports a black-box payload across two connections, mechanical retention and regulated power, and the operator commands the aircraft on a separate radio path. (b) At 10 km separation the ideal screen blocks the direct path and clears the relay path.

Table 2. Failure consequences and evidence needed for the four flows

Flow	Allocated element	Consequence if lost	Next evidence
Mission traffic	Relay payload	Relay service is lost	Installed two-hop delivery test
Aircraft command	Carrier avionics	Aircraft control is at risk	Command continuity during a bounded payload fault
Electrical power	Carrier distribution	Relay dwell or carrier control is at risk	Payload-rail load and fault-protection test
Mechanical support	Carrier mount	Payload retention is at risk	Fit, load, and vibration checks

Protection thresholds, allowable interruptions, and the carrier's response set the pass criterion for that test.

Electrical power. The carrier battery supplies propulsion, avionics, and a regulated payload branch, so the carrier owns the power flow up to the payload connector. The carrier analysis assumes a continuous 14 W payload load behind a regulator with 0.92 efficiency, and it shows how that load and the 25 W auxiliary allowance affect dwell and gross mass. This 14 W value is a sizing assumption for the study rather than an electrical interface specification. Rail voltage, peak current, connector, protection, and thermal behavior follow from the payload radio selection, and a load test across that voltage and current envelope then verifies the rail.

Mechanical support. The carrier mount retains the payload and transfers its load into the airframe. The model includes the 0.20 kg payload and a mount allowance of 0.20 kg plus 18 percent of payload mass. Detailed design must specify the mount geometry, load envelope, retention criterion, and inspection access. The candidate accommodation requirement REQ-012, listed with the verification plan in Section 3, addresses fit, mass, and volume. Retention needs an explicit requirement and supporting inspection and load-test evidence before flight.

This allocation lets the designer change placement without changing the assumed payload installation. Changing the radio is a different decision, since similar mass and power do

not establish interchangeability; mounting, voltage, thermal behavior, and electromagnetic compatibility must also match. Overall, each flow in Table 2 sits at one of three evidence levels in this paper. Mission traffic and the mass and power consequences of the carrier flows are screened by calculation. Aircraft command and payload retention are allocations with proposed interface tests. The remaining attributes stay open until a radio is selected. This makes Table 2 an interface-development record that shows which evidence each flow needs next.

Study Measures and Candidate Criteria

The study represents a ground operator seeking an outbound path to a remote UAS across an obstruction. Two candidate measures of effectiveness (MOEs) describe the service: separation at a specified dwell, and dwell at a specified separation. Link margin, gross mass, and rotor size provide the technical measures and constraints used to screen those targets. Following the distinction in the NASA handbook [9], gross mass and installed hover power are also candidates for technical performance measures (TPMs) once a specific design is tracked through development.

Table 3 retains the analyst-selected comparison points from the existing grid. The 10 km, 30-minute case is the reference decision; 20 km and 45 minutes separately increase the requested reach and dwell. This arrangement makes the trade easy to interpret. The 15 kg mass ceiling and 0.75 m rotor limit define the study's size class. Acceptance values for the candidate requirements REQ-001, REQ-010, and REQ-011 come from the Stakeholder Expectations step in Figure 2. A pass therefore means that a candidate survives the declared screen.

Table 3. Candidate service measures for the notional study

Measure and held condition	Threshold	Objective
MOE-1: separation at 30 min, 120 m relay height	10 km	20 km
MOE-2: dwell at 10 km, 120 m relay height	30 min	45 min

Analyst-selected comparison points from the sampled grid.

3. METHOD

Figure 2 shows the four system design processes of the NASA handbook [9], and each part of the study belongs to one of them. Stakeholder Expectations become the outbound relay need and the candidate service measures in Table 3. Requirements Definition produces the candidate requirements traced in Table 6. Logical Decomposition becomes the carrier-payload boundary and four-flow allocation in Section 2. Design Solution Definition is where the link and carrier screens evaluate each candidate against its success criteria. This means every result in Section 4 sits at a known point in the process. When a screen fails, the loop in Figure 2 identifies which earlier step the designer returns to.

The study workflow is summarized in Figure 3: define the service, allocate responsibilities, declare the inputs, and screen the link and carrier separately. The baseline grid holds placement at the midpoint and varies separation, height, and dwell. A supplementary analytical comparison moves the relay toward the ground endpoint while retaining the same radio, payload, height, and carrier assumptions. It uses the same path-loss relation and explicitly checks screen clearance. That comparison is separate from the saved 90-case midpoint grid and does not change its counts.

For a fixed payload, the two screens are separable: dwell does not enter the link calculation, and separation and altitude do not change the constant-density hover model. Their intersection is a joint feasibility screen. The carrier model itself retains the battery-mass-power feedback. The decision rule is to carry forward a candidate that meets the declared target, examine which sensitivities change that result, and specify evidence for the remaining uncertainties. The independent checker addresses implementation consistency; published hover data calibrate the model's power term.

Declared Inputs

The study samples declared geometries and durations rather than observed operations. Heights are measured from a common ground datum, with the ground endpoint at 0 m and the remote aircraft at 100 m. The relay sits at the horizontal midpoint at 60, 120, or 240 m. Endpoint separation takes six values from 5 to 60 km and dwell takes five values from 10 to 60 minutes. Table 4 lists these inputs. The baseline scenario places an 80 m opaque screen at 40 percent of the separation. Four variants change the screen's height or position, one variant adds 12 dB of loss, and one comparator removes the screen entirely. These are simple visibility cases chosen to isolate geometry.

The payload values come from a commercial mesh radio module with a published mass of 102 g, 5 W receive consumption, 14 W peak consumption, and 1.6 W maximum transmit power [10]. The model adds 98 g for unselected integration hardware and, as a sizing assumption, applies the 14 W peak value continuously. The model treats the stated maximum RF output and DC consumption as a simultaneous operating point. The receiver threshold of -90 dBm is one common assumption applied to every receiver, and the 5 MHz bandwidth is recorded as context that never enters the margin calculation.

Each scenario holds 90 cases, formed from six separations, five dwell values, and three relay altitudes. The primary payload is the 0.20 kg, 14 W case. Seven scenarios and three payload cases give 1,890 saved records. Since these records repeat the same grid instead of describing independent missions, every count in this paper uses the 90-case denominator.

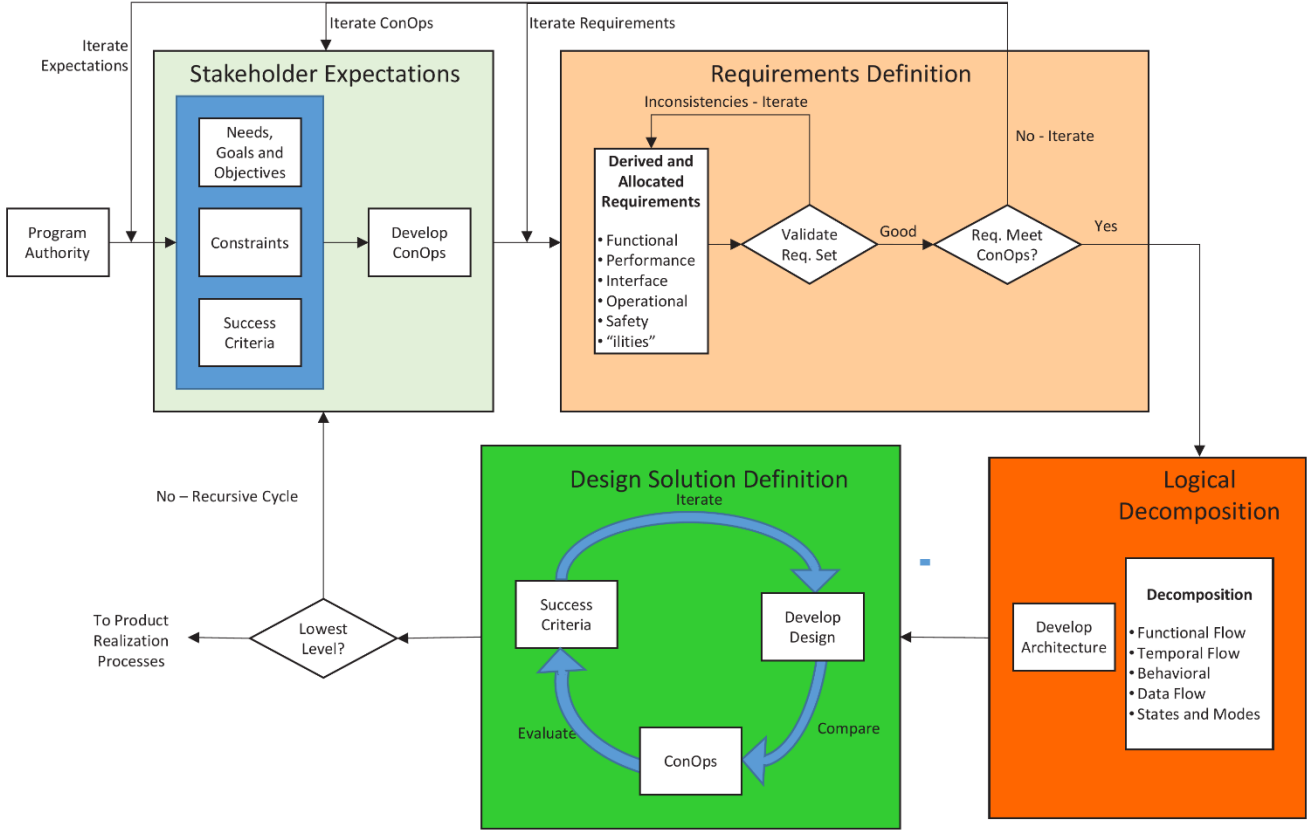


Figure 2. NASA system design processes, reproduced from [9, Fig. 4.0-1]. Section 2 carries out Stakeholder Expectations and Logical Decomposition for the relay, Table 6 records Requirements Definition, and the link and carrier screens perform Design Solution Definition.

Link Check

The link check asks two questions about each path: is it clear, and is the received signal strong enough? A path is clear when the straight line between its endpoints passes above the screen. By similar triangles, the relay height at which the first hop begins to clear depends on the screen height and its position along the path.

For the baseline 80 m screen at 40 percent of the separation and a midpoint relay, this threshold is 100 m. This means a relay at or below 100 m leaves the first hop blocked. Heights are compared strictly, so a path that just grazes the screen counts as blocked. In every screened scenario the screen sits on the first hop, so the second hop does not cross it. The screen is modeled as an opaque plane over flat earth, and Section 5 examines how diffraction and Fresnel-zone clearance would move the threshold.

For a clear path, the check computes free-space loss from distance and frequency with the standard transmission relation [11], [12].

The link margin is the received level above the receiver threshold,

$$M = P_t + G_t + G_r - L - L_0 - L_x - S, \quad (1)$$

where L is the free-space loss. Here P_t is the transmit level in dBm, G_t and G_r are antenna gains in dBi, and S is the -90 dBm threshold. L_0 is a 6 dB allowance for other losses, and L_x is the scenario loss. The ground endpoint transmits at 30 dBm and the relay transmits at 32.04 dBm, which is the 1.6 W maximum expressed in dBm. A relay path passes only when both hops are clear and both margins are at least 0 dB.

Carrier Sizing

The aircraft side of the assessment asks whether an aircraft of consistent mass can hover with the payload for the requested time. Rotor disk area is set by holding disk loading constant at 60 N/m^2 , so area grows in proportion to mass. Momentum theory then gives ideal hover power, and dividing by the rotor figure of merit and the drive efficiency converts that to electrical power [13], [14]. The result is proportional to gross mass m ,

$$P = km, \quad k = \frac{g k_{env}}{FM \eta} \sqrt{\frac{DL}{2\rho}}, \quad (2)$$

where FM is the rotor figure of merit, η the drive efficiency, DL the disk loading, ρ the air density, and k_{env} a multiplier for operating conditions. The calibration identifies only the combined term $FM \eta/k_{env}$, so the reference relation reproduces laboratory hover power with no margin for operating conditions. At the calibrated values in Table 4, k is 108.4 W/kg. This means the model describes a family of aircraft that are resized as mass grows.

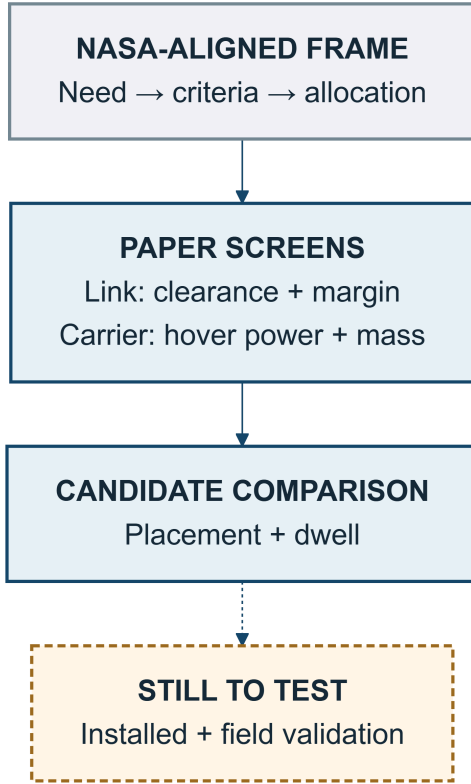


Figure 3. Study workflow using selected NASA system design practices [9]. Gray summarizes the analyst-defined need, candidate criteria, and functional allocation. Blue shows this paper's link and carrier screens and candidate comparison. Installed verification and operational validation remain outstanding.

The battery must satisfy two separate demands, and the larger one governs. The energy demand covers hover power plus a fixed 40.2 W for avionics and the payload over the full dwell, using only the usable fraction of the pack. The power demand covers the peak current that a 1.75 thrust ratio implies, with a further 15 percent allowance. Dividing each demand by the pack's specific energy or specific power gives a battery mass, and Appendix A states both relations.

Holding back 20 percent after a 90 percent depth of discharge leaves a usable fraction of 0.72. This reserve is part of the mission policy rather than the calibration.

The reference screen includes a 20 percent battery reserve, a 1.75 thrust ratio, a 15 percent peak-power allowance, and 6 dB of other link loss. These are separate allowances, and one cannot be traded for another. In particular, the hover

calibration absorbs the declared operating-condition multiplier and supplies no additional operating-power margin. A separate 15 percent hover-power stress case tests that omission. The allowances must ultimately be justified against the chosen equipment, mission, and acceptance criteria.

Table 4. Declared inputs and calibrated coefficients

Quantity	Value	Status
Endpoint separation	5, 10, 20, 30, 40, 60 km	Declared grid
Dwell	10, 20, 30, 45, 60 min	Declared grid
Ground, remote height	0, 100 m	Common datum
Relay position; height	midpoint; 60, 120, 240 m	Not optimized
Screen height; position	80 m; 0.40 of span	Declared scenario
Frequency; bandwidth	2400 MHz; 5 MHz	Bandwidth unused
Transmit levels	30 dBm ground; 32.04 dBm relay	Declared
Antenna gains	6 dBi endpoint; 2 dBi relay	Declared
Other loss; threshold	6 dB; −90 dBm	Declared
Extra loss	0 dB; 12 dB case	Declared scenario
Payload mass; DC power	0.20 kg; 14 W	Device data plus allowance
Effective FM parameter	0.6203	Aggregate fit [15]
Structure intercept	0.163 kg	Anchored to sized design [7]
Battery specific energy e	192.1 Wh/kg	Production pack [16]
Drive efficiency η	0.83	Declared
Disk loading DL	60 N/m ²	Declared
Condition multiplier k_{env}	1.15	Declared; absorbed by fit
Battery specific power p	900 W/kg	Declared
Usable fraction u	0.72	Mission policy
Thrust margin	1.75	Declared
Fixed carried mass F	1.116 kg	Declared allowances
Structure growth fraction	0.18	Declared
Size limits	15 kg; 0.75 m rotor	Screening limits

Declared values are analysis assumptions rather than approved requirements. The aggregate hover fit, structural anchor, and battery reference are explained below. The appendix gives all mass-model coefficients.

Adding the remaining masses closes the loop. Propulsion mass scales with peak power and rotor area, and structure carries a base mass plus a fraction of everything it supports. The payload, avionics, power electronics, and mount contribute a fixed 1.116 kg, and Appendix A defines each allowance. Since every term is either fixed or proportional to gross mass, each battery branch reduces to a fixed mass plus a growth factor times the trial mass.

Table 5. Hover calibration and held-out endurance check

Vehicle or test article	Role	Mass (kg)	Rotor dia. (m)	Battery (Wh)	Published	Model	Error
3DR SOLO, NASA hover test [15]	fit	1.51	0.254		184.4 W	180.7 W	-2.0%
DJI Phantom 3, NASA hover test [15]	fit	1.22	0.239		172.1 W	140.1 W	-18.6%
3DR Iris+, NASA hover test [15]	fit	1.28	0.244		142.9 W	146.6 W	+2.6%
Drone America DAX8, NASA hover test [15]	fit	5.59	0.279		838.0 W	827.5 W	-1.3%
SUI Endurance, NASA hover test [15]	fit	2.91	0.381		281.5 W	322.7 W	+14.6%
DJI Matrice 30 [16]	check	3.77	0.406	263.2	36.0 min	33.6 min	-6.7%
DJI Matrice 4 [18]	check	1.22	0.274	99.5	42.0 min	40.8 min	-2.9%
DJI Mavic 3M [19]	check	0.95	0.239	77.0	37.0 min	38.2 min	+3.2%

Fit rows compare measured electrical hover power at the test article's own thrust and rotor size. Check rows were held out of the fit and compare published hover endurance at the manufacturer's stated discharge condition, which does not include the 20 percent mission reserve used elsewhere in this paper. The DAX8 and Endurance were hover-tested inside the wind-tunnel test section, where recirculation was not quantified; the other three NASA articles were tested in a laboratory about 30 ft from the nearest wall [17]. The DJI Matrice 350 was excluded because its published 55 minutes is forward flight at about 8 m/s rather than hover. The rotor diameters for the three check aircraft are derived from the published propeller designation rather than quoted directly.

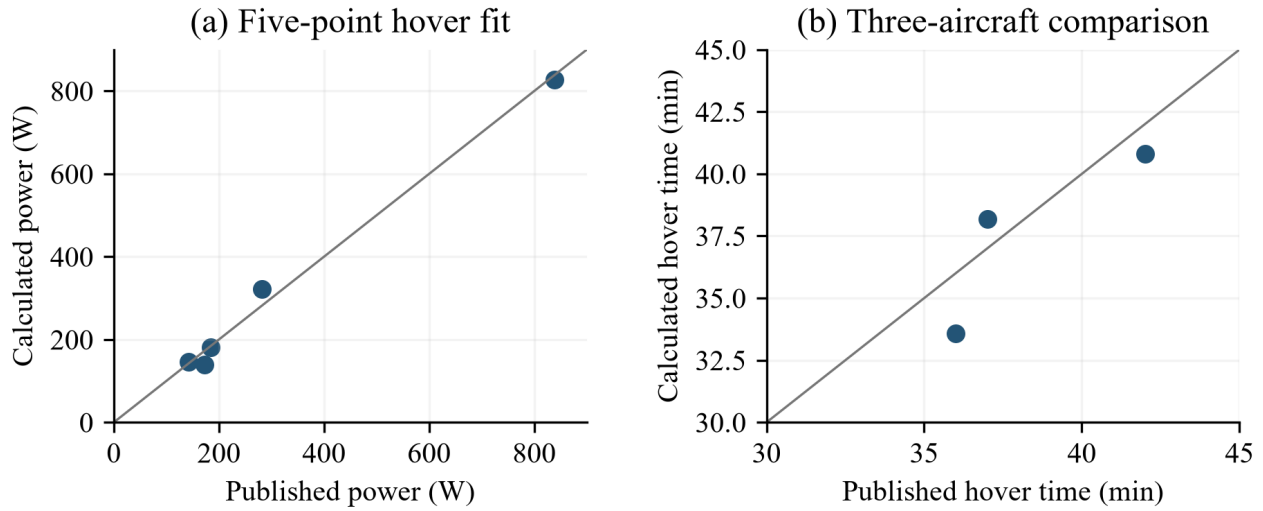


Figure 4. Hover evidence shown on separate axes. (a) Measured and fitted electrical powers for the five NASA points. (b) Published and calculated maximum-hover times for three aircraft excluded from the fit. A diagonal indicates agreement; neither panel validates carrier mass closure.

The growth factor is the fraction of any added kilogram that the aircraft must add again as battery, propulsion, and structure. The energy branch grows with dwell, while the power branch does not. At each trial mass the model takes the larger battery requirement.

When the Aircraft Closes

A consistent sizing result is a fixed point, where the mass implied by the resource calculation equals the trial gross mass. Each branch gives a candidate equal to its fixed mass divided by one minus its growth factor, and Appendix A states the tests a candidate must pass to count as a solution.

The denominator explains why a longer dwell can have a disproportionate mass penalty. As the governing slope approaches one, a small increase in demand produces a large increase in closed mass. Once the energy slope reaches one, its positive intercept prevents a finite positive fixed point. This is a property of the assumed model. It may not hold as a general endurance limit for multirotors.

At the reference inputs, power governs the 10- and 20-minute examples, while energy governs the 30- and 45-minute examples. For the architecture, this means the carrier's limiting resource shifts from delivered power to stored energy as the requested dwell grows. The implementation checks this analytical test with a separate iterative solution, described in Appendix A.

Calibration and Check

The hover calibration estimates one aggregate coefficient from five hover test points reported in the 2018 NASA technical memorandum [15]. For each point, ideal induced power uses the measured supported thrust and the sum of that test article's nominal rotor swept areas. Measured electrical power is voltage multiplied by the sum of motor currents. An unweighted least-squares fit through the origin relates measured electrical power to ideal induced power. The fit identifies the combined effect of rotor, drive, and operating conditions rather than an independently measured rotor figure of merit. Its residuals range from -18.6 to +14.6

percent. The Phantom 3 point is the largest disagreement: the fitted relation gives 140.1 W against a published measurement of 172.1 W. That point is kept in the calibration set.

Because the fit passes through the origin, each point's influence scales with the square of its ideal power; the DAx8 carries 79 percent of that weight. The two highest-leverage observations were also collected inside the wind-tunnel test section, as the 2016 AHS Forum paper describes [17]. To expose this dependence, the analysis also refits the coefficient to each test point alone. These single-point refits measure sensitivity to the selected calibration point and give no statistical bound on endurance; Section 4 reports the boundaries they produce. Appendix B examines other estimators and documents the DAx8 eight-disk area definition against the source photographs.

The structural intercept is inferred separately from the NASA conceptual-design mass breakdown [7]. That breakdown describes a sized baseline design rather than a weighed vehicle. The calculation reproduces its airframe allowance after subtracting the present model's structural growth and disk-area terms, and several carried masses in that subtraction are model allowances. The resulting 0.163 kg is therefore conditional on those choices. The reference specific energy, 192 Wh/kg, comes from the published TB30 pack energy divided by its mass [16]. The 900 W/kg continuous power capability is declared separately. The 192 Wh/kg value is also the lowest pack value among the check aircraft. The Matrice 4 pack stores 99.5 Wh in 401 g, or 248 Wh/kg [18], and Section 4 uses that value as a battery sensitivity case.

Three commercial quadrotors are excluded from the five-point fit and used for a separate hover-time comparison [16], [18], [19]. Their own mass, rotor area, and nominal battery energy enter the calibrated power relation, with a common assumed 25 W auxiliary load. The calculation uses the published maximum-hover discharge condition without the relay mission's depth-of-discharge and reserve reductions. Differences range from -6.7 to +3.2 percent (Table 5 and Figure 4). The auxiliary allowance matters for the two smaller aircraft. Only the Matrice 30, whose hover power is large relative to the allowance, stays within 10 percent across a 0 to 40 W sweep (Appendix B). The two smaller aircraft are therefore consistent with the fitted relation rather than an independent test of it. The Matrice 350 figure is excluded because it describes forward flight. These comparisons provide a consistency check on hover power and energy accounting.

Size Limits and Decision Order

A closed aircraft is screened against two size limits that keep the result in the small relay class. Gross mass must not exceed 15 kg, and the equivalent rotor diameter must not exceed 0.75 m. The diameter comes from the total disk area shared equally across four rotors.

At the reference disk loading the rotor limit binds first, since a 0.75 m rotor corresponds to a gross mass of 10.81 kg. A

separate span check is twice the rotor diameter, so it repeats the rotor limit rather than adding a new one. These limits define the relay class under study.

Each case then receives one label from the first check that fails. If the direct path passes the assumed service screen, the case is DIRECT: relay benefit is not required by this criterion. Otherwise, if either relay hop is blocked or short of margin, the case is LINK FAIL. Otherwise, if the aircraft has no solution, the case is NO SOLUTION, and if it has a solution outside the size limits, the case is TOO LARGE. What remains is a PASS, meaning a blocked or insufficient direct path, two passing outbound relay hops, and a finite carrier within the exploratory limits. Because connectivity is checked first, a LINK FAIL label can hide an aircraft failure in the same case, so the case records keep both flags.

Verification and Validation Plan

The evidence separates implementation checks from physical comparisons and planned installation tests. The documented equations are evaluated by scripted analysis code that generates every case record. That code carries 35 automated tests, and a separate checker rebuilds each link from coordinates, solves the mass balance by independent bisection, and compares its results with the saved records. This shows that the implementation follows the equations. Five published hover observations calibrate one aggregate coefficient. Commercial-aircraft hover times provide an external consistency check, with the auxiliary-load sensitivity described above.

Verification asks whether a product meets its specified requirements; validation asks whether it serves its intended use [9]. Early analysis and review can support those activities without a flight-ready aircraft. This study sits at the Design Solution Definition step of Figure 2, before the service criteria are agreed with an operator. The modeled passes therefore support design selection, and compliance belongs to the product realization processes that follow. Before hardware acceptance, each relevant criterion must name the configuration, conditions, observable result, and acceptable threshold.

Table 6 records the existing evidence and planned checks using analysis, inspection, demonstration, and test. It is the planning matrix for the next phase. Every flow in Table 2 traces to at least one candidate requirement: REQ-001 to mission traffic, REQ-004 to aircraft command, REQ-015 to electrical power, and REQ-012 to mechanical support. REQ-010 and REQ-011 carry the carrier resource limits, and REQ-016 constrains the interface itself. The IDs follow the full candidate requirement set, so the gaps in numbering belong to requirements for return traffic, transit, and other mission phases outside this assessment boundary. This means each allocated flow has a named requirement and a planned verification method before hardware is selected. For the candidate carried forward, the key checks are the two installed radio hops, hover power with the payload fitted, and command continuity under a bounded payload fault. Their acceptance thresholds come from the agreed service criteria.

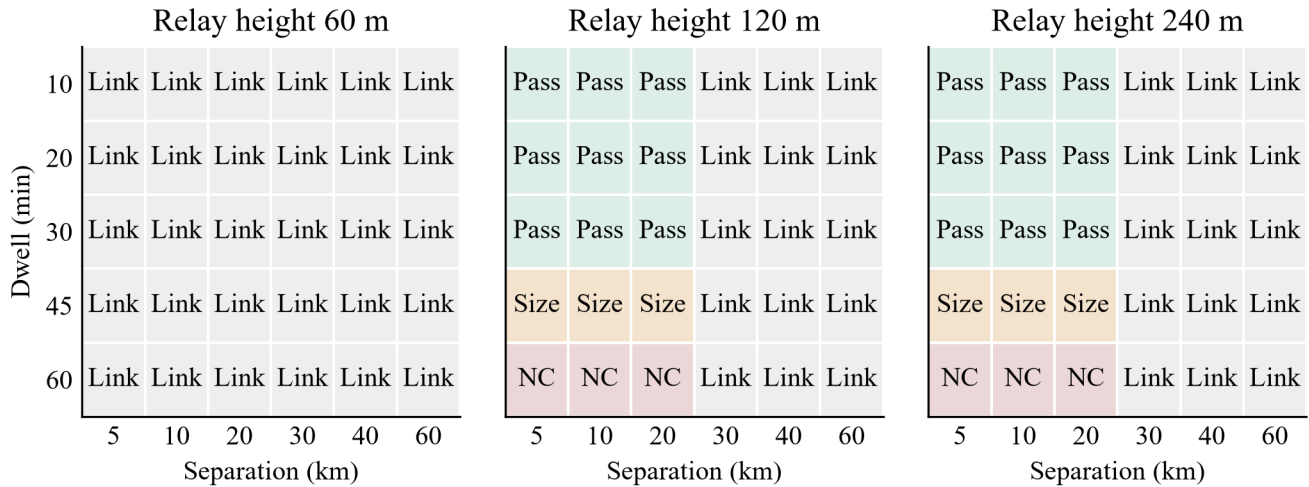


Figure 5. Baseline results for the 90 primary cases. A PASS means the relay helps and the aircraft stays within the screening limits, SIZE marks a solution outside those limits, NC marks a case with no solution, and LINK FAIL marks the earlier link check. All cells in the 60 m panel fail visibility. Link failures in the 120 m and 240 m panels are clear but short of margin. No case in the baseline is direct-sufficient.

Table 6. Candidate requirement trace and verification plan

Candidate	Allocated function or constraint	Current evidence	Planned verification and acceptance basis
REQ-001	Relay outbound traffic	Geometry and received-level analysis only	Test: two-hop communication with the payload installed; owner-selected delivery criterion remains open
REQ-004	Accept aircraft commands	Allocation only; no evidence yet	Demonstration: command and health path with payload traffic present, absent, and faulted, showing that a payload fault does not interrupt command; fault indication, carrier response, and control criterion remain open
REQ-010	On-station endurance	Coupled hover mass closure for declared dwells	Test: timed hover with the payload installed and the mission reserve held; owner dwell target remains open
REQ-011	Limit gross mass	Model mass and exploratory size screens	Test: weigh the configured aircraft against an owner-approved mass ceiling
REQ-012	Accommodate payload	Assumed 0.20 kg mass; volume and mount loads not modeled	Inspection: check installed payload fit against the selected mass and volume envelope; mount load path and retention not yet specified
REQ-015	Regulated payload power	Study assumption of continuous 14 W; not an interface specification; rail voltage, peak current, connector, protection, and thermal limits open	Test: load the rail across its specified voltage and current envelope once a payload radio is selected
REQ-016	Limit carrier-payload crossings	Model review of power and mechanical allocation	Inspection: check that the as-built interface has only the two specified crossings

An operational trial would then assess the agreed service with an operator at a representative site, including the mission phases omitted from this model.

4. RESULTS

Where the Envelope Ends

These results complete the Design Solution Definition step of Figure 2. The baseline supports further assessment of the 10 km, 30-minute target, but not the 45-minute objective with the reference carrier assumptions. Figure 5 shows the midpoint grid, and the link and carrier boundaries explain its

labels. At 120 m, the midpoint relay reaches a calculated 25.0 km separation before the ground-to-relay hop runs out of margin. Adding 12 dB of loss reduces that boundary to 6.3 km, below even the 10 km target. More battery cannot repair this failure while the radio assumptions stay fixed. The designer must reconsider placement, radio performance, or the requested separation. The carrier boundaries below answer a different question: how long a radio path that passes can be supported.

The reference carrier reaches the 0.75 m rotor limit at 42.1 minutes. At that point gross mass is 10.81 kg, so the rotor limit is tighter than the 15 kg mass ceiling. Finite closure

persists until the energy-branch slope reaches one at 52.4 minutes. This means the interval between these boundaries holds aircraft that close mathematically but fail the physical screen. Figure 6 shows this transition without treating the divergent branch as a buildable aircraft.

The existing favorable and adverse parameter sets illustrate dependence on carrier assumptions. The favorable set gives a 126.9-minute physical boundary and a 157.0-minute energy-slope crossing. The adverse set has no physical pass even at zero dwell; its 6.4-minute energy-slope crossing is not an achievable service duration. These sets change several assumptions simultaneously. They are deterministic corners rather than a confidence interval.

The decision should not use 42.1 minutes as available endurance. Refitting the hover coefficient to each test point alone moves the rotor boundary from 33.0 to 49.2 minutes and the energy-slope crossing from 41.2 to 61.3 minutes. A separate 15 percent hover-power allowance moves the reference boundaries to 35.7 and 44.5 minutes. Since both stresses act through the power coefficient k in (2), they compound when applied together. The least favorable refit is the Phantom 3 point. Pairing it with the 15 percent allowance places the rotor boundary near 28 minutes. At 30 minutes, that carrier closes only at about 16 kg with a 0.92 m rotor. The other four refits paired with the allowance keep the rotor boundary between about 35 and 42 minutes. This means the 30-minute target passes each stress applied alone but fails the least favorable pairing. Therefore, the study carries 30 minutes forward as a candidate that installed hover-power measurement must confirm, and it defers any 45-minute commitment.

Substituting 248 Wh/kg while holding the other battery assumptions fixed moves the reference boundaries to 54.4 and 67.7 minutes. Adding battery to the same design raises the mass that the battery must lift. Therefore, a longer dwell would need a justified change to the carrier itself, such as a qualified higher-energy pack or lower installed hover power.

One Geometry, Three Outcomes

The worked case fixes the geometry and varies only dwell, which isolates the aircraft behavior. At 10 km separation the screen stands 4 km from the ground endpoint. The direct line passes it at 40 m, below the 80 m top, so the direct path is blocked. With the relay at 120 m the first hop passes the screen at 96 m and is clear, as Figure 1(b) shows. The two hop margins are 7.97 and 10.02 dB, and they do not change with dwell. This is what makes the case useful: every difference that follows belongs to the aircraft. The saved record also lists a 5.96 dB direct margin, which is the margin the direct path would have if the screen were not there.

Table 7 shows what changes instead. At 10 and 20 minutes the battery is sized by the power it has to deliver rather than the energy it has to store. This means both cases close at the same 3.58 kg, and the installed 231 Wh is more than either dwell needs. At 30 minutes energy takes over, the aircraft closes at 4.77 kg with a 0.50 m rotor, and the battery reaches

42 percent of gross mass. At 45 minutes the aircraft still closes, but only at 15.15 kg with a 0.89 m rotor, which fails both size limits. At 60 minutes each added kilogram of battery calls for more than another kilogram to lift it, so there is no aircraft to report at all. The last two rows are the two failure stages from Section 3 seen in one geometry.

The short-dwell result also needs a scale check. Adding the assumed 0.20 kg payload to the published masses gives 3.97 kg for the Matrice 30 and 1.42 kg for the Matrice 4. Under the fitted power relation and stated energy policy, both yield about 22 minutes, compared with the 3.58 kg, 20-minute reference case. These are calculations only. The reference family's 1.116 kg fixed allowance approaches the 1.22 kg Matrice 4 mass in Table 5, and its short-dwell pack is power-sized. That bookkeeping helps explain the difference. Selecting an actual airframe remains the next step for the 30-minute candidate.

Table 7. Worked case at 10 km separation and 120 m relay height

Dwell (min)	Mass (kg)	Rotor (m)	Pack sized by	Energy factor a	Result
10	3.58	0.43	power	0.346	PASS
20	3.58	0.43	power	0.500	PASS
30	4.77	0.50	energy	0.654	PASS
45	15.15	0.89	energy	0.885	TOO LARGE
60	none	none	none	1.116	NO SOLUTION

Hop margins are 7.97 and 10.02 dB throughout. The energy-branch factor is tabulated; the power-branch factor is 0.570. A pass requires relay benefit, two passing hops, and compliance with both size limits.

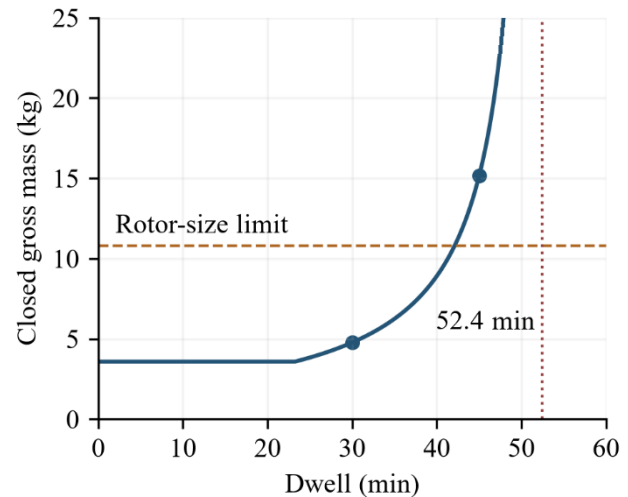


Figure 6. Reference carrier mass versus dwell, evaluated from the stated branch equations. The rotor screen binds at 42.1 minutes; the energy slope reaches one at 52.4 minutes. The plotted finite curve is clipped at 25 kg for readability.

The Sampled Grid

The 90-case grid shows where those two limits fall across the sampled points. Of the 90 baseline cases, 18 pass, 6 are too large, 6 have no solution, and 60 fail the link check. Figure 5 shows the pattern. Every case at 60 m fails because the relay sits below the 100 m clearance threshold. At 120 and 240 m

the 5, 10, and 20 km cases clear the screen and hold margin. Within those cases, dwells from 10 to 30 minutes pass, the 45-minute cases are too large, and the 60-minute cases have no solution. At 30 km and beyond the relay can see both endpoints but the first hop falls short of margin, which matches the 25.0 km limit above.

Of the 90 cases, 30 pass the link check and 54 pass the carrier check; 18 pass both. Table 8 retains the joint failures hidden by the ordered labels. Because the screens depend on separate grid coordinates, the value of their intersection is diagnostic. Of the 60 link failures, 24 also fail the carrier screen, so improving only the radio path would not recover them. This is why the assessment keeps both flags for every case. While link failures dominate this selected grid, both the 10 km threshold and the 20 km objective pass nominally at 30 minutes and 120 m.

Table 8. Joint link and carrier screens for 90 baseline cases

	Carrier pass	Carrier fail
Link pass	18	12
Link fail	36	24

The clear comparator shows why relay benefit has to be judged against the direct path. With the screen removed, the 5 and 10 km direct paths carry positive margin, so 30 cases are labeled DIRECT. Only 9 cases, all at 20 km, are relay passes, because at 20 km the direct margin is -0.07 dB while both relay hops stay positive. The relay helps in a narrow band: close enough that two hops hold margin, far enough that one hop does not.

Scenario Sensitivity

Adding 12 dB of loss leaves 6 passes, 2 too large, 2 with no solution, and 80 link failures. Only the 5 km cases retain both hop margins; at 10 km the first hop falls to -4.03 dB. Carrier results remain unchanged. This stress case reverses the nominal 10 km service decision, so site-specific link evidence matters before committing to that target.

Moving the screen changes the first-hop clearance threshold and therefore changes which altitude layers work. Figure 7 shows the five thresholds. Lowering the screen to 60 m or moving it to 0.45 of the span drops the threshold to 75.0 and 88.9 m. This leaves the sampled results unchanged, since 60 m is still blocked and 120 m is still clear. Raising the screen to 100 m or moving it to 0.30 of the span raises the threshold to 125.0 and 133.3 m. This blocks the 120 m layer and halves the passes to 9. While the two lower thresholds produce no change at all, this reflects the sampled altitudes. No altitude between 75 and 120 m was tested, so the result may still be sensitive in that range. For siting, the relay height has to be chosen against the actual obstruction, because a screen that raises the threshold above 120 m removes that altitude layer from service.

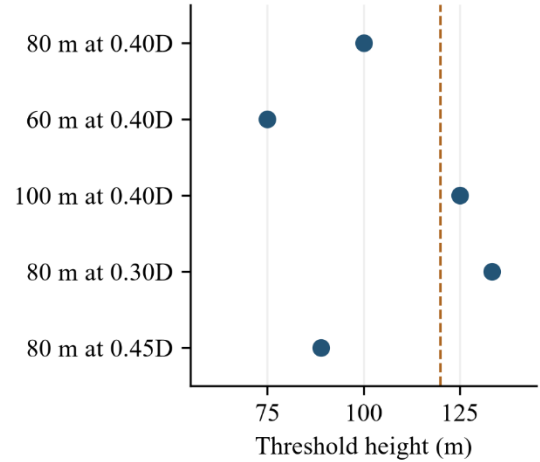


Figure 7. Geometric clearance thresholds; values above 120 m exclude that relay height.

The two secondary payload cases preserve the primary classification boundaries in the baseline grid. Mass, DC demand, and RF output change together, so this is a comparison of declared payload cases rather than an isolated mass or power experiment.

Placement Choice for the 30-Minute Candidate

The midpoint is a useful baseline, but it need not be the preferred site. Under the declared asymmetric transmit powers, shifting the relay toward G balances the two hop margins at approximately 0.442 of the endpoint separation. This follows by finding each hop's allowable slant distance at zero margin, converting it to horizontal distance at the stated heights, and placing the relay in proportion to those distances. The rounded position describes a candidate for site assessment.

Table 9 compares the two placements at 10 km separation, 120 m height, and 30 minutes. The midpoint leaves about 10.0 dB on the second hop but only 7.97 dB on the first, while the offset leaves about 9.0 dB on each, and both first hops clear the 80 m screen. The calculated separation boundary rises from 25.0 km to about 28 km.

Both positions retain the reference 4.77 kg carrier and 0.50 m rotor because placement does not enter the hover model. The roughly 1 dB gain on the weaker hop is useful because it costs nothing in the carrier screen. However, it is modest beside the 6 dB other-loss allowance and cannot absorb the 12 dB stress loss. The offset is therefore carried forward provisionally for site assessment, with the midpoint retained as the reproducibility baseline. A real siting decision must also account for access, terrain, antenna patterns, command coverage, and the energy needed to reach and leave the position.

Table 9. Placement at 10 km and 120 m relay height

Measure	Unit	Midpoint	Offset
Carrier mass	kg	4.77	4.77
First-hop height at screen	m	96.0	108.6
Weaker-hop margin	dB	7.97	9.04

5. DISCUSSION

What to Carry Forward

The evidence supports carrying the 10 km, 30-minute target forward, with a relay near 0.442 of the span at 120 m as the candidate for site assessment and the midpoint case as the comparison baseline. While the offset improves the weaker hop under the declared geometry, the carrier result is the less certain half of the decision. The 30-minute target survives each hover stress applied alone, but the least favorable calibration point combined with the 15 percent power allowance moves the rotor boundary below it. Installed hover-power measurement therefore bears directly on whether the dwell target holds. This makes hardware selection and confirmation of the intended service with an operator the next decisions.

The present carrier evidence does not support a 45-minute commitment. Its nominal 15.15 kg result already exceeds both size limits, and the calibration sensitivity changes whether that dwell is supportable. At 60 minutes the reference relations have no finite mass solution. These are different failures. The first asks for a change in size, resources, or demand, while the second requires a change in the scaling assumptions or demand before resizing can help. Relaxing the mass ceiling therefore cannot repair mathematical nonclosure.

The useful systems result is a diagnosis tied to an action. Each failed screen points back to one of the four flows. A weak radio path is a mission-traffic problem and calls for a siting or radio decision. An excessive carrier is a power and mechanical-support problem and calls for a dwell or resource decision. A payload fault that can disturb aircraft command sits where the power and command flows meet, and it calls for an interface decision. This is the practical value of the allocation. In Figure 2 terms, a siting or radio decision returns to Design Solution Definition, a dwell decision returns to Requirements Definition, and an interface decision returns to Logical Decomposition. A radio upgrade, for example, would need its RF benefit, mass, DC load, antenna installation, and compatibility assessed together. A gain in one screen is not automatically a gain for the complete relay.

The next evidence should address whichever uncertainty can reverse the intended decision. Agreeing the service with the operational user comes first, including return traffic, delivery criteria, dwell, and reserve policy, since those criteria define what a compatible payload-carrier candidate must satisfy. Where site loss is poorly known, installed measurements of both hops at the midpoint and offset placements would test the link half of the recommendation. Where dwell drives selection, repeated installed hover-power measurements that record mass, battery state, and operating conditions would test the carrier half, and a later energy budget would add climb, transit, and recovery. Where command independence

is essential, the bounded payload-fault assessment described in Section 2 would test the power and command interface. Each test needs acceptance criteria defined before it runs. These activities can proceed in parallel.

Limitations

The carrier model describes a resized family at constant disk loading. It does not represent an existing aircraft that can swap batteries. Its aggregate hover fit does not resolve scale effects, rotor interference, or environmental variation. The DAX8 dominates the fit, and the two highest-leverage points were measured in the tunnel with unquantified recirculation. The single-point range and the combined pairings with the 15 percent allowance are sensitivity results, and they do not bound all model error. Other pairings of a calibration stress with an individual battery or structural assumption were not examined. The commercial hover comparisons use nominal battery energy and an assumed auxiliary load, so small discrepancies may include compensating assumptions.

The structural relation is anchored at one conceptual mass breakdown. Its extrapolation becomes especially consequential near the closure singularity. Neither that anchor nor the power fit establishes packaging, strength, vibration response, thermal compatibility, or a realizable combination of battery energy and power.

The link calculation tests visibility and received level. It omits interference, fading, diffraction, Fresnel clearance, scheduling, duplex operation, and throughput. These simplifications do not all err in the same direction. An opaque screen treats a grazing path as blocked, whereas knife-edge diffraction would impose a finite loss. However, omitting Fresnel clearance works in the other direction just above the geometric threshold. In the 10 km case, keeping 60 percent of the first Fresnel zone clear raises the 100 m threshold to about 108 m, which the 120 m layer still clears.

The offset-placement comparison is limited to the same ideal screen and fixed antenna gains as the baseline. A more favorable path in this geometry need not be a better installed site: terrain, antenna patterns, interference, command coverage, and mission energy may change the choice. Statistical air-to-ground blockage models address different questions [20]; none is fitted or validated here. Neither the placement result nor the count of grid passes supplies a probability of service availability.

Only stationary outbound dwell is calculated. Transit, climb, recovery energy, return traffic, carrier-command performance, including the carrier's response to a payload fault, and station keeping remain outside the quantitative assessment. A follow-on assessment would need matched power measurements, installed-payload characterization, a complete mission energy budget, and communications measurements for the intended service.

6. CONCLUSIONS

For the declared 10 km outbound service, the study carries a 30-minute dwell target forward for detailed assessment. It also identifies a relay near 0.442 of the span at 120 m as a candidate for site evaluation. The reference carrier closes at 4.77 kg with a 0.50 m rotor, and the offset improves the weaker hop by about 1 dB without changing modeled hover demand. The 45-minute reference case exceeds both size limits. However, the 30-minute target does not survive every stress examined, since pairing the least favorable calibration point with a 15 percent hover-power allowance places the rotor boundary near 28 minutes. The nominal 42.1-minute boundary is therefore a model result, and installed hover-power measurement is what turns it into an endurance figure.

The contribution is the connection, made through the four-flow allocation, between a failed screen and the next engineering decision. Placement can improve a radio path but cannot cure excessive carrier mass, and additional battery cannot recover a blocked or inadequate link. Before committing to hardware, the service criteria and payload-carrier interfaces must be baselined. Site-specific link measurements, installed hover-power measurements, and a bounded payload-fault assessment address different ways the recommendation could fail. Full-mission energy and operational validation follow in the product realization phase. Overall, following the NASA system design processes turned each failed screen into a named return step. The result is a candidate worth testing, with a stated reason for each test.

APPENDICES

A. COMPLETE CARRIER MASS BALANCE

The following equations specify the component bookkeeping used in the results. Mass is in kilograms, power in watts, energy in watt-hours, area in square meters, and dwell t in hours. Write $q = g/DL$, so total rotor disk area is $A = qm$, and let k denote electrical propulsion power per unit mass from (2). Define

$$P_0 = P_{\text{aux}} + P_L/\eta_r, \quad u = d_b(1 - r_b). \quad (3)$$

The reference values are $P_0 = 40.2174$ W and $u = 0.72$. With thrust ratio τ and battery power margin μ_b , the battery branches are

$$\begin{aligned} m_E &= \alpha_E m + \beta_E, & \alpha_E &= kt/(ue), \\ \beta_E &= P_0 t/(ue), & \alpha_P &= \mu_b \tau^{3/2} k/p, \\ m_P &= \alpha_P m + \beta_P, & \beta_P &= \mu_b P_0/p. \end{aligned} \quad (4)$$

Battery mass is $m_b = \max(m_E, m_P)$. Installed nominal energy is em_b ; on the power branch it can exceed the nominal energy required for dwell. The model applies thrust scaling to propulsion alone.

With the component coefficients in Table 10, fixed carried mass and propulsion-group mass are

$$\begin{aligned} m_{\text{mount}} &= m_{m0} + f_m m_L, \\ F &= m_L + m_{\text{av}} + m_{\text{pe}} + m_{\text{mount}}, \\ m_{\text{prop}} &= K_p m, \quad K_p = \tau^{3/2} k/p_p + \sigma_r q. \end{aligned} \quad (5)$$

The structural allowance includes all carried mass:

$$m_s = m_{s0} + f_s (F + m_b + m_{\text{prop}}) + \sigma_s A. \quad (6)$$

Substitution in $m_{\text{new}} = F + m_b + m_{\text{prop}} + m_s$ gives, for branch $j \in \{E, P\}$,

$$\begin{aligned} a_j &= (1 + f_s)(\alpha_j + K_p) + \sigma_s q, \\ b_j &= (1 + f_s)(F + \beta_j) + m_{s0}. \end{aligned} \quad (7)$$

Table 10. Remaining reference carrier coefficients

Quantity	Value
Gravity g ; density ρ	9.80665 m/s ² ; 1.225 kg/m ³
Rotor count n	4
Propulsion specific power p_p	2300 W/kg
Rotor area mass σ_r	0.14 kg/m ²
Structure fraction f_s	0.18
Structure area mass σ_s	0.22 kg/m ²
Avionics mass m_{av}	0.40 kg
Power electronics m_{pe}	0.28 kg
Mount base m_{m0} ; fraction f_m	0.20 kg; 0.18
Auxiliary power P_{aux}	25 W
Payload regulator η_r	0.92
Depth of discharge d_b	0.90
Reserve fraction r_b	0.20
Thrust ratio τ ; margin μ_b	1.75; 1.15

Evaluate $m_j^* = b_j/(1 - a_j)$ only for $a_j < 1$ and positive mass. The energy candidate is valid when $m_E(m_E^*) \geq m_P(m_E^*)$; the power candidate requires the reverse inequality at m_P^* . With no valid candidate, physical mass, area, power, and energy are undefined. A branch tie gives the same fixed point.

For reproduction, the reference effective FM is 0.620274, structure intercept is 0.163173 kg, and specific energy is 192.117 Wh/kg. Other nominal values appear in Table 4. These numerical precisions identify inputs rather than measurement accuracy. The 1.5 m span proxy equals twice rotor diameter and repeats the 0.75 m rotor screen. The full input catalog supplies the simultaneous favorable and adverse parameter sets; those corners are not used for the primary 90-case counts.

The implementation also solves the mass balance by relaxed iteration as a diagnostic, starting at 5 kg with a 200-iteration limit. The analytical branch test controls all reported physical quantities, so a timeout or the 50 kg iteration guard cannot be mistaken for physical nonclosure.

B. HOVER CALIBRATION SENSITIVITY

The single-point refits in Section 3 describe sensitivity to the calibration observations; they do not bound all hover-model uncertainty. The Phantom 3 and Endurance points, whose disk loadings of 67 and 63 N/m² are nearest the reference value, set both ends of that range. Other defensible estimators (mean ratio, geometric mean, quadrotors only) give 41.4 to

44.6 minutes for the rotor limit and 51.5 to 55.6 minutes for the energy-slope crossing. This means the choice of estimator moves the boundaries less than the choice of test point.

The DAX8 area uses eight 11 in rotor disks. The 2016 AHS Forum paper lists eight rotors of that diameter, and its test-article photographs show eight laterally separated rotor positions rather than four coaxial pairs [17, Table 1 and Figs. 5 and 12]. Their summed geometric area is approximately 0.490 m². The reference calibration therefore retains the eight-disk convention. A four-disk substitution does not represent the documented arrangement and is not used to support a longer dwell. This resolves the geometry question. However, the limits of the calibration remain, since rotor interference is still embedded in the aggregate coefficient and the DAX8 still dominates the fit. The single-point and alternative-estimator checks remain the relevant tests of calibration sensitivity.

The hover-time comparison in Section 3 is also sensitive to the assumed auxiliary load. With a 0 W allowance, the Matrice 30, Matrice 4, and Mavic 3M differences become -1.5, +17.2, and +30.0 percent. With 10 W they become -3.7, +8.2, and +17.8 percent, and with 40 W they become -9.6, -11.9, and -8.2 percent. Only the Matrice 30 remains within 10 percent across this sweep; its calculated discrepancy still depends on the allowance.

DATA AVAILABILITY

The computational baseline, declared inputs, calibration data, 1,890 saved cases, tests, and independent checker are archived at <https://github.com/noahstephenson/low-cost-relay-uas> under the tag `aeroconf-2027-paper-2784-r3`, which covers the midpoint grid, calibration, and checks. The offset-placement comparison and the combined hover-stress pairings follow directly from the equations and inputs reported here, so they can be reproduced from that baseline.

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The views expressed in this paper are those of the author. They do not reflect the official policy or position of the United States Military Academy, the Department of the Army, the Department of Defense, or the U.S. Government.

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BIOGRAPHY



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